Common Distribution Types: Part I

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1 Introduction

Data distributions indicate how values are spread across in a dataset. By quickly spotting common distribution types, you can easily determine certain qualities and patterns the dataset contains (*not me sounding like some marketing click-bait).* Anyway, let's begin!

2 Discrete Distributions

Countable, separate numbers (ie values from countable set).

- **Bernoulli:** single trial data with only 2 possible outcomes where $X \in \{0, 1\}$.
  - Ex: 1 coin toss
- **Binomial:** the number of successes from fixed number of independent n-Bernoulli trials with constant p.
  - Ex: 3 rounds of coin tosses where success is heads
- **Poisson:** number of independent events occurring within a fixed interval at constant average rate λ
  - Ex: you dropped your phone 5 times from noon to 3pm
- **Geometric:** how many trials until your first success, long right tail when success is uncommon.
  - Ex: number of job applications sent until you get a job

3 Continuous Distributions

Measurable numbers that may include fractions/decimals on a scale (ie values across continuous range).

- **Normal:** classic bell-shaped curve with majority of data in the middle, values common near mean while extreme deviations are rare
  - Ex: congrats you are extremely average height within the homogeneous population
- **Uniform:** constant probability density across continuous interval (ie equal density) and no region is favored
  - Ex: `np.random.uniform(0,5)` randomly samples numbers from $[0, 5)$.
- **Exponential:** amount of time that passes between independent (Poisson-type) events where $rate = \lambda$ and $N(t) \sim \text{Poisson}(\lambda t)$ and waiting time $T \sim \text{Exponential}(\lambda)$, generally right-skewed where short times are common and long wait times are rare.
  - Ex: the amount of time between each phone drop bc you are just so clumsy

4 Skews

- **Right-skewed:** positive skew with a long tail on the right side (ie most values are low).
- **Left-skewed:** negative skew with a long tail on the left side (ie most values are high).
- **Bimodal:** distribution contains 2 identifiable peaks (contains 2 different modes). Consider the possibility of mixed population or other underlying processes
- **Multi-modal:** bro wtf did you do.

5 References

1. NIST, Gallery of Distributions
2. Wikipedia, List of probability distributions